Siva Surisetty

ce24mtech02002@iith.ac.in | sivakalikiran123@gmail.com | +91 7658924580 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	9.19
Government Polytechnic College Kancharapalem	Class XII (State Board of Technical Education and Training (SBTET))	2020	91.17%
Ramanath Secondary School	Class X (Central Board of Secondary Education (CBSE))	2017	8.80%

RESEARCH EXPERIENCE

M.Tech (RA) Thesis: Integrated Forecasting Framework for Reservoir Inflow & Operation | *Jan 2024* - *Present*

1. Benchmarked AR, ARX, MLR, ANN, LSTM, GRU and wavelet hybrids (WMLR, WLSTM) for 1 to 3 day daily inflow forecasts
2. Evaluated atmospheric, land surface and flow predictors to find the best input and model combination per lead time
3. Developed monthly inflow forecasts with LSTM and graph neural network hybrids (GCN-LSTM, GAT-LSTM)
4. Implemented the HBV model with real-time data assimilation using Kalman Filter and Ensemble Kalman Filter
5. Delivered real-time inflow forecasts up to 7 days ahead using NWP inputs (GEFS, GFS, ECMWF IFS)
6. Developing Forecast Informed Reservoir Operation (FIRO) to optimise releases for demand, flood and drought control

PUBLICATIONS

"Intercomparison of input information and model structures for daily reservoir inflow forecasting across lead times", *Journal of Hydrologic Engineering* (ASCE), under review (submitted Sep 2026)

PROJECTS

Real-Time Reservoir Inflow Forecasting Web Application | *Sponsored research project*

- Built a full-stack, login-gated decision-support tool delivering NWP-driven inflow forecasts up to 7 days ahead
- Automated NWP data ingestion (GEFS, GFS, ECMWF IFS), model runs and interactive map/chart visualisation

LiDAR-Based Riparian Zone Monitoring | *Riverathon 1.0 (NMCG & World Bank)* | *Feb 2026*

- Developed a geospatial workflow to monitor riparian vegetation from LiDAR data to support river restoration

Satellite Precipitation Downscaling & Merging (SRF-DM) | *Course project* | *Mar 2024*

- Built a Random-Forest-based spatial downscaling and merging method fusing satellite, reanalysis and gauge precipitation

SCHOLASTIC ACHIEVEMENTS

1. Winner, 1st Prize, **Riverathon 1.0**, a national river hackathon by NMCG (Ministry of Jal Shakti), supported by the World Bank; 1st among 320 teams (~1,800 participants) | *Feb 2026*
2. Recipient, **Research Appreciation Award 2026**, IIT Hyderabad | *Apr 2026*

WORK EXPERIENCE

Project Assistant, IIT Hyderabad | *Jun 2023 - Dec 2023*

TECHNICAL SKILLS

1. **Programming:** Python, R, MATLAB, TypeScript
2. **ML / Forecasting:** Time-series forecasting, LSTM, GRU, ANN, GCN-LSTM, GAT-LSTM, wavelet hybrids, AR/ARX/MLR, Random Forest; TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Polars
3. **Hydrology & Data Assimilation:** HBV, HEC-HMS, Kalman Filter, Ensemble Kalman Filter
4. **Web & Data Engineering:** FastAPI, React, Vite, DuckDB, GeoPandas, MapLibre GL, ECharts, Git
5. **Geospatial & HPC:** QGIS, Google Earth Engine, LiDAR processing, HPC (Slurm, parallel computing)

POSITIONS OF RESPONSIBILITY

1. **Teaching Assistant**, Hydraulic Engineering Lab, IIT Hyderabad | *Jul - Dec 2026*
2. **Coordinator**, HydroMasteR Workshop (ML for hydrology), IIT Hyderabad | *Dec 2024*

TRAININGS & CERTIFICATIONS

1. Hydrological Analysis & Operation of Reservoir Systems, NIH Roorkee (Ministry of Jal Shakti) | *Jul 2025*
2. Applied Machine Learning using Python (AML-Py'25), NIT Kurukshetra | *Jan 2025*